



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Fall 2023

Paper: Calculus-II

Course Code: CAL-212

Roll No. 016122

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(6×5=30)

Q.1. Solve the following:

- (i) Find two positive numbers x and y whose sum is 30 and are such that xy^2 is as large as possible.
- (ii) For what value of x in the interval $-1 \leq x \leq 4$, is the graph of the function $f(x) = 2x^2 - \frac{x^3}{3}$ steepest? What is the slope of tangent at this point?
- (iii) A business manager determines that t months after production begins on a next product, number of units produced will be p million per month, where $p = \frac{t}{(t+1)^2}$. Find $p'(t)$ & $p''(t)$. What happen to the production in the long run as $t \rightarrow \infty$?
- (iv) Find the area of the region enclosed by the line $y=4x$ and the curve $y = x^3 + 3x^2$.
- (v) Evaluate the definite integral $\int_1^4 \left(x^{\frac{3}{2}} + \frac{2}{x} \right) dx$.
- (vi) After t hours on job, a factory worker can produce $100te^{-0.5t}$ units per hour. How many units does the worker produce during the first 3 hours?

Solve the following:

(3×10=30)

Q. 2 Evaluate the integrals: (a) $\int (\ln x)^2 dx$ (b) $\int \frac{x+2}{e^{3x}} dx$.

Q. 3 A company produces x units of commodity A and y units of commodity B . All the units can be sold for $p=100-x$ dollars per unit of A and $q=100-y$ dollars per unit of B . The cost in dollars of producing these units is given by the function $C(x, y) = x^2 + xy + y^2$. What should be x and y to maximize the profit?

Q. 4 Find the rate of change of x as $\frac{dy}{dx}$ for the prescribed value of x :

(a) $y = (x^2 + 2)(x + \sqrt{x})$, $x = 4$.

(b) $y = \frac{x-1}{x^2+x+1}$, $x = 1.5$.